Bid Document

General

This package contains the contextual and assembly model for the function defined in the root class.

Bid contextual model

General

This package contains all the artefacts for the Bid Contextual Document.

!# shows the diagram.

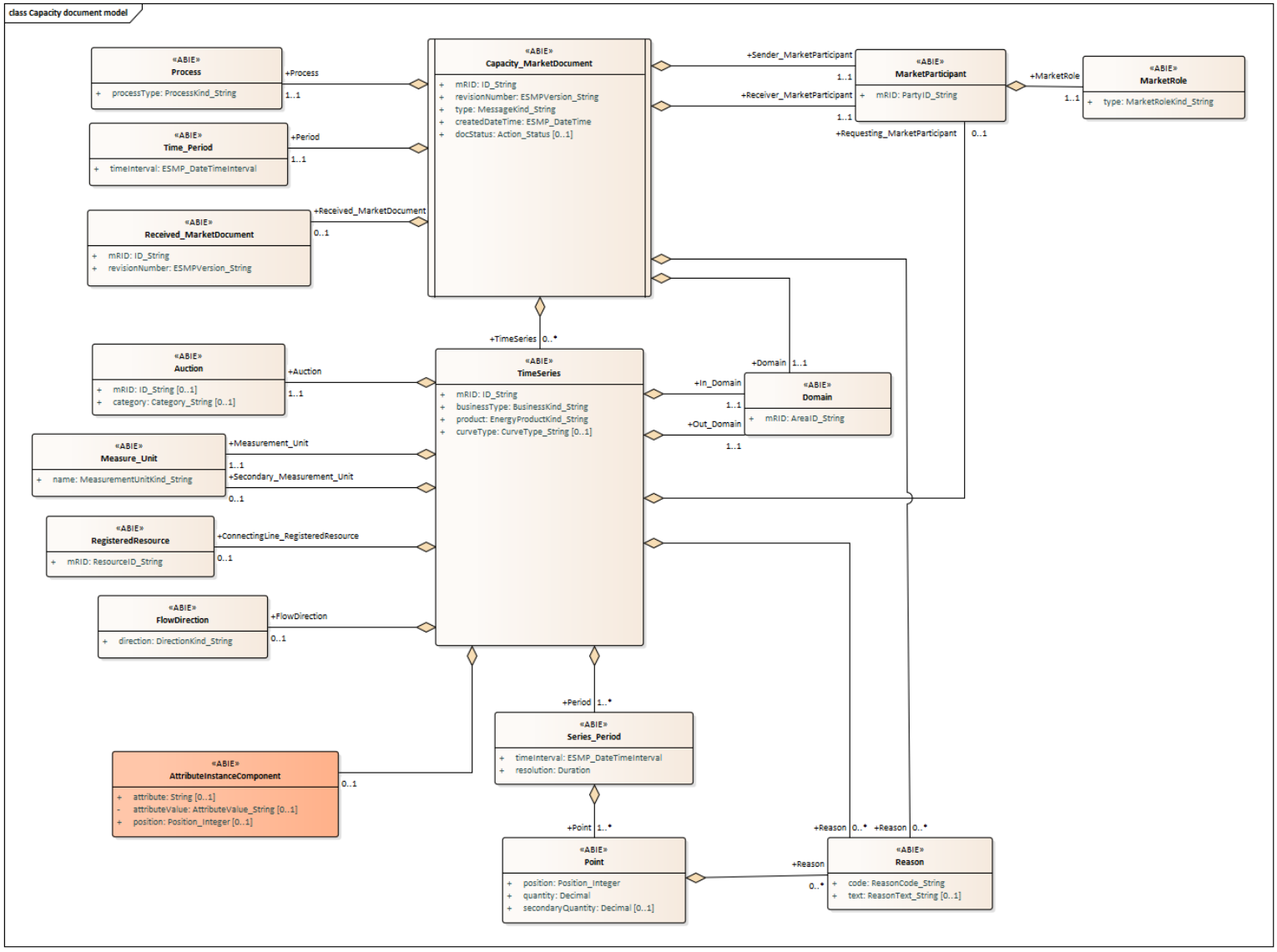


Figure !# - diagram Capacity bid document model

(ABIE) Auction

The identification of a formal specification of an energy product that is offered for sale.

!# shows all attributes of Auction.

Table !# - Attributes of Bid contextual model::Auction

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID String	The unique identification of the auction.

(ABIE) Bid_MarketDocument

A bid document contains a set of bids (a bid is represented by a time series). There may be several bids submitted by the sender for the same bid period and subject party.
An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of Bid_MarketDocument.

Table !# - Attributes of Bid contextual model::Bid_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID String	The unique identification of the document being exchanged within a business process flow.
revisionNumber (ABIE)	1..1	ESMPVersion String	The identification of the version that distinguishes one evolution of a document from another.
type (ABIE)	1..1	MessageKind String	The coded type of a document. The document type describes the principal characteristic of the document.
createdDateTime (ABIE)	1..1	ESMP_DateTime	The date and time of the creation of the document.

!# shows all association ends of Bid_MarketDocument with other classes.

Table !# - Association ends of Bid contextual model::Bid_MarketDocument with other classes

mult.	mult.	Class type name	Description
Domain (ABIE)	[1..1]	Domain	The domain covered within the bid document, i.e. the border for which auction is done. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Domain.Domain[0..1]
Period (ABIE)	[1..1]	Time_Period	The beginning and ending date and time of the period covered by the document. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Time_Period.Period[0..*]
Subject_MarketParticipant (ABIE)	[1..1]	MarketParticipant	The party for whom the bid is being submitted. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]

Receiver_MarketParticipant (ABIE)	[1..1]	MarketParticipant	Document recipient. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]
Sender_MarketParticipant (ABIE)	[1..1]	MarketParticipant	Document owner. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: MarketParticipant.MarketParticipant[0..*]

(ABIE) BidTimeSeries

The formal specification of specific characteristics related to a bid.

!# shows all attributes of BidTimeSeries.

Table !# - Attributes of Bid contextual model::BidTimeSeries

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID String	A unique identification of the time series.
businessType (ABIE)	1..1	BusinessKind String	The identification of the nature of the time series.
divisible (ABIE)	1..1	ESMPBoolean String	An indication whether or not each element of the bid may be partially accepted or not.
linkedBidsIdentification (ABIE)	0..1	ID String	The unique identification used to identify associated bids with each other.
blockBid (ABIE)	1..1	ESMPBoolean String	The indication that the values in the period are considered as a whole. They cannot be changed or subdivided.

!# shows all association ends of BidTimeSeries with other classes.

Table !# - Association ends of Bid contextual model::BidTimeSeries with other classes

mult.	mult.	Class type name	Description
Out_Domain (ABIE)	[1..1]	Domain	The area where the energy is coming from. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Domain.Domain[0..*]
Price_Measurement_Unit (ABIE)	[0..1]	Measure_Unit	The unit of measure in which the price in the time series is expressed (MW, MWh, etc.). Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Measure_Unit.Measurement_Unit[0..*]

Quantity_Measurement_Unit (ABIE)	[1..1]	Measure_Unit	The unit of measure in which the quantities in the time series are expressed, e.g. MAW. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Measure_Unit.Measurement_Unit[0..*]
In_Domain (ABIE)	[1..1]	Domain	The area where the energy is to be put. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Domain.Domain[0..*]
Currency_Unit (ABIE)	[0..1]	Currency_Unit	The currency in which the monetary amount is expressed. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Currency_Unit.Currency_Unit[0..1]
Auction (ABIE)	[1..1]	Auction	The identification linking the bid to a set of specifications created by the auction operator. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Auction.Auction[0..*]

(ABIE) Currency_Unit

The code specifying a monetary unit.

!# shows all attributes of Currency_Unit.

Table !# - Attributes of Bid contextual model::Currency_Unit

Attribute name	mult.	Attribute type	Description
name (ABIE)	1..1	CurrencyCode_String	The identification of the formal code for a currency (ISO 4217).

(ABIE) Domain

A domain covering a number of related objects, such as market balance area, grid area, borders etc.

!# shows all attributes of Domain.

Table !# - Attributes of Bid contextual model::Domain

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	AreaID_String	The unique identification of the domain.

(ABIE) MarketParticipant

The identification of the party participating in energy market business processes.

!# shows all attributes of MarketParticipant.

Table !# - Attributes of Bid contextual model::MarketParticipant

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	PartyID String	The identification of a party in the energy market.

!# shows all association ends of MarketParticipant with other classes.

Table !# - Association ends of Bid contextual model::MarketParticipant with other classes

mult.	mult.	Class type name	Description
MarketRole (ABIE)	[1..1]	MarketRole	Association Based On : ESMPClasses:: MarketParticipant.[] ----- ESMPClasses:: MarketRole.MarketRole[0..1]

(ABIE) MarketRole

The identification of the intended behaviour of a market participant played within a given business process.

!# shows all attributes of MarketRole.

Table !# - Attributes of Bid contextual model::MarketRole

Attribute name	mult.	Attribute type	Description
type (ABIE)	1..1	MarketRoleKind String	The identification of the role played by a market player.

(ABIE) Measure_Unit

A particular quantity, defined and adopted by convention, with which other quantities of the same kind are compared in order to express their magnitudes relative to that quantity.

!# shows all attributes of Measure_Unit.

Table !# - Attributes of Bid contextual model::Measure_Unit

Attribute name	mult.	Attribute type	Description
name (ABIE)	1..1	MeasurementUnitKind String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20).

(ABIE) Point

The quantity that is bid for the interval in question.
The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of Bid contextual model::Point

Attribute name	mult.	Attribute type	Description
position (ABIE)	1..1	Position Integer	A sequential value representing the relative position within a given time interval.
quantity (ABIE)	1..1	Decimal	The principal quantity identified for a point.

!# shows all association ends of Point with other classes.

Table !# - Association ends of Bid contextual model::Point with other classes

mult.	mult.	Class type name	Description
Price (ABIE)	[0..1]	Price	The price expressed for each unit of quantity. The price amount is mandatory in the case of capacity auctions and shall not be provided in the case of rule based allocations depending on local market rules (for example "first come first serve"). Association Based On : ESMPClasses:: Point.[] ----- ESMPClasses:: Price.Price[0..*]

(ABIE) Price

The cost corresponding to a specific entity expressed in a currency.

!# shows all attributes of Price.

Table !# - Attributes of Bid contextual model::Price

Attribute name	mult.	Attribute type	Description
amount (ABIE)	1..1	Amount Decimal	A number of monetary units specified in a unit of currency.

(ABIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of Bid contextual model::Series_Period

Attribute name	mult.	Attribute type	Description
timeInterval (ABIE)	1..1	ESMP DateTimeInterval	The start and end time of the period.
resolution (ABIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.

(ABIE) Time_Period

The identification of a time interval.

!# shows all attributes of Time_Period.

Table !# - Attributes of Bid contextual model::Time_Period

Attribute name	mult.	Attribute type	Description
timeInterval (ABIE)	1..1	ESMP DateTimeInterval	The start and end date and time for a given interval.

Bid assembly model

General

This package contains the model that has been assembled from the contextual model. All classes with relations [1..1] or [0..1] have had their attributes integrated into their parent class.

!# shows the diagram.

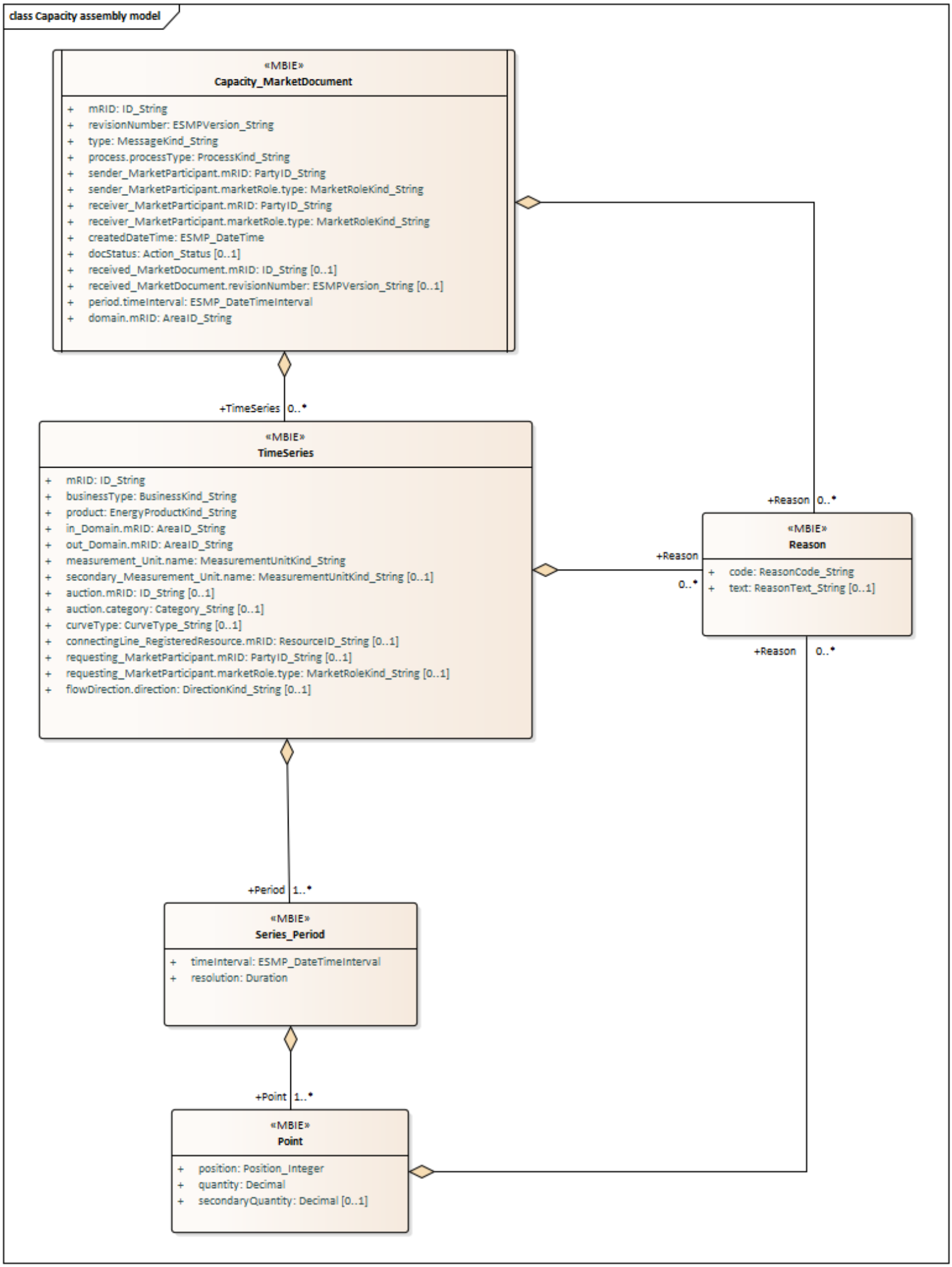

class Capacity assembly model
«MBIE»
Capacity_MarketDocument
+ mRID: ID_String
+ revisionNumber: ESMPVersion_String
+ type: MessageKind_String
+ process.processType: ProcessKind_String
+ sender_MarketParticipant.mRID: PartyID_String
+ sender_MarketParticipant.marketRole.type: MarketRoleKind_String
+ receiver_MarketParticipant.mRID: PartyID_String
+ receiver_MarketParticipant.marketRole.type: MarketRoleKind_String
+ createdDateTime: ESMP_DateTime
+ docStatus: Action_Status [0..1]
+ received_MarketDocument.mRID: ID_String [0..1]
+ received_MarketDocument.revisionNumber: ESMPVersion_String [0..1]
+ period.timeInterval: ESMP_DateTimeInterval
+ domain.mRID: AreaID_String
+TimeSeries
0..*
«MBIE»
TimeSeries
+ mRID: ID_String
+ businessType: BusinessKind_String
+ product: EnergyProductKind_String
+ in_Domain.mRID: AreaID_String
+ out_Domain.mRID: AreaID_String
+ measurement_Unit.name: MeasurementUnitKind_String
+ secondary_Measurement_Unit.name: MeasurementUnitKind_String [0..1]
+ auction.mRID: ID_String [0..1]
+ auction.category: Category_String [0..1]
+ curveType: CurveType_String [0..1]
+ connectingLine_RegisteredResource.mRID: ResourceID_String [0..1]
+ requesting_MarketParticipant.mRID: PartyID_String [0..1]
+ requesting_MarketParticipant.marketRole.type: MarketRoleKind_String [0..1]
+ flowDirection.direction: DirectionKind_String [0..1]
+Reason
0..*
+Reason
0..*
«MBIE»
Reason
+ code: ReasonCode_String
+ text: ReasonText_String [0..1]
+Reason
0..*
+Period
1..*
«MBIE»
Series_Period
+ timeInterval: ESMP_DateTimeInterval
+ resolution: Duration
+Point
1..*
«MBIE»
Point
+ position: Position_Integer
+ quantity: Decimal
+ secondaryQuantity: Decimal [0..1]

Figure !# - diagram Bid assembly model

(MBIE) Bid_MarketDocument

A bid document contains a set of bids (a bid is represented by a time series). There may be several bids submitted by the sender for the same bid period and subject party.

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of Bid_MarketDocument.

Table !# - Attributes of Bid assembly model::Bid_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	1..1	ID String	The unique identification of the document being exchanged within a business process flow.
revisionNumber (MBIE)	1..1	ESMPVersion String	The identification of the version that distinguishes one evolution of a document from another.
type (MBIE)	1..1	MessageKind String	The coded type of a document. The document type describes the principal characteristic of the document.
sender_MarketParticipant.mRID (MBIE)	1..1	PartyID String	The identification of a party in the energy market. --- Document owner.
sender_MarketParticipant.marketRole.type (MBIE)	1..1	MarketRoleKind String	The identification of the role played by a market player. --- Document owner.
receiver_MarketParticipant.mRID (MBIE)	1..1	PartyID String	The identification of a party in the energy market. --- Document recipient.
receiver_MarketParticipant.marketRole.type (MBIE)	1..1	MarketRoleKind String	The identification of the role played by a market player. --- Document recipient.
createdDateTime (MBIE)	1..1	ESMP DateTime	The date and time of the creation of the document.
			The start and end date and time for a given interval.

period.timeInterval (MBIE)	1..1 ESMP DateTimeInterval	--- The beginning and ending date and time of the period covered by the document.
domain.mRID (MBIE)	1..1 AreaID String	The unique identification of the domain. --- The domain covered within the bid document, i.e. the border for which auction is done.
subject_MarketParticipant.mRID (MBIE)	1..1 PartyID String	The identification of a party in the energy market. --- The party for whom the bid is being submitted.
subject_MarketParticipant.marketRole.type (MBIE)	1..1 MarketRoleKind String	The identification of the role played by a market player. --- The party for whom the bid is being submitted.

(MBIE) BidTimeSeries

The formal specification of specific characteristics related to a bid.

!# shows all attributes of BidTimeSeries.

Table !# - Attributes of Bid assembly model::BidTimeSeries

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	1..1	ID String	A unique identification of the time series.
auction.mRID (MBIE)	1..1	ID String	The unique identification of the auction. --- The identification linking the bid to a set of specifications created by the auction operator.
businessType (MBIE)	1..1	BusinessKind String	The identification of the

			nature of the time series.
in_Domain.mRID (MBIE)	1..1	AreaID String	The unique identification of the domain. --- The area where the energy is to be put.
out_Domain.mRID (MBIE)	1..1	AreaID String	The unique identification of the domain. --- The area where the energy is coming from.
quantity_Measurement_Unit.name (MBIE)	1..1	MeasurementUnitKind String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure in which the quantities in the time series are expressed, e.g. MAW.
currency_Unit.name (MBIE)	0..1	CurrencyCode String	The identification of the formal code for a currency (ISO 4217). --- The currency in which the monetary amount is expressed.
price_Measurement_Unit.name (MBIE)	0..1	MeasurementUnitKind String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure in which the price in the time series is expressed (MW, MWh, etc.).
divisible (MBIE)	1..1	ESMPBoolean String	An indication whether or not each element of the bid may be partially accepted or not.
linkedBidsIdentification (MBIE)	0..1	ID String	The unique identification used to identify associated bids with each other.
blockBid (MBIE)	1..1	ESMPBoolean String	The indication that the values in the period are considered as a whole.

They cannot be changed or subdivided.

(MBIE) Point

The quantity that is bid for the interval in question.
The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of Bid assembly model::Point

Attribute name	mult.	Attribute type	Description
position (MBIE)	1..1	Position Integer	A sequential value representing the relative position within a given time interval.
quantity (MBIE)	1..1	Decimal	The principal quantity identified for a point. A number of monetary units specified in a unit of currency.
price.amount (MBIE)	0..1	Amount Decimal	--- The price expressed for each unit of quantity. The price amount is mandatory in the case of capacity auctions and shall not be provided in the case of rule based allocations depending on local market rules (for example "first come first serve").

(MBIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of Bid assembly model::Series_Period

Attribute name	mult.	Attribute type	Description
timeInterval (MBIE)	1..1	ESMP DateTimeInterval	The start and end time of the period.
resolution (MBIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.

Global Used Datatypes

Datatypes

ID_String

A code to uniquely distinguish one occurrence of an entity from another.

In the ESMP context, the code is defined either by:

- an emitting company that provides an agreed identification unique within a business context such as capacity auction identification, market agreement identification, etc.

- a party (originator of the exchange) that provides a unique identification in the framework of a business exchange such as document identification, time series identification, bid identification, ...

!# shows all attributes of ID_String.

Table !# - Attributes of ESMPDataTypes::ID_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

ESMPVersion_String

In ESMP, the coded value is restricted to digits.

A code that distinguishes one evolution of an identified object from another. Information about a specific object may be sent several times, each transmission being identified by a different version number.

!# shows all attributes of ESMPVersion_String.

Table !# - Attributes of ESMPDataTypes::ESMPVersion_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

MessageKind_String

The coded type of a document.

!# shows all attributes of MessageKind_String.

Table !# - Attributes of ESMPDataTypes::MessageKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	MessageTypeList	Main Core value Space.

ESMP_DateTime

In ESMP, the dateTime shall be expressed in UTC as YYYY-MM-DDThh:mm:ssZ.

!# shows all attributes of ESMP_DateTime.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	Main Core value Space.

BusinessKind_String

The coded identification of the business type.

!# shows all attributes of BusinessKind_String.

Table !# - Attributes of ESMPDataTypes::BusinessKind_String.

Attribute name	mult.	Attribute type	Description

value (CIMDatatype)	1..1	BusinessTypeList	Main Core value Space.

ESMPBoolean_String

The attribute is a Boolean, either 0 or 1 value, "No" or "Yes", or two values defined in a code list.

!# shows all attributes of ESMPBoolean_String.

Table !# - Attributes of ESMPDataTypes::ESMPBoolean_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	IndicatorTypeList	Main Core value Space.

CurrencyCode_String

The coded identification of legal tender using ISO 4217 3 alpha codes.

!# shows all attributes of CurrencyCode_String.

Table !# - Attributes of ESMPDataTypes::CurrencyCode_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	CurrencyTypeList	Main Core value Space.

AreaID_String

The coded identification of a domain, i.e. balance area, grid area, etc.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for domain identification.

!# shows all attributes of AreaID_String.

Table !# - Attributes of ESMPDataTypes::AreaID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

PartyID_String

The identification of an actor in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for market participant identification.

!# shows all attributes of PartyID_String.

Table !# - Attributes of ESMPDataTypes::PartyID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

MarketRoleKind_String

The identification of the role played by a party.

!# shows all attributes of MarketRoleKind_String.

Table !# - Attributes of ESMPDataTypes::MarketRoleKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	RoleTypeList	Main Core value Space.

MeasurementUnitKind_String

The coded identification of a unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

!# shows all attributes of MeasurementUnitKind_String.

Table !# - Attributes of ESMPDataTypes::MeasurementUnitKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	UnitOfMeasureTypeList	Main Core value Space.

Position_Integer

An integer value, this value is used as a sequential value representing the relative position of an entity within a space such as a time interval.

!# shows all attributes of Position_Integer.

Table !# - Attributes of ESMPDataTypes::Position_Integer.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	Integer	Main Core value Space.

Amount_Decimal

The coded identification of a monetary value.

!# shows all attributes of Amount_Decimal.

Table !# - Attributes of ESMPDataTypes::Amount_Decimal.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	Decimal	Main Core value Space.

YMDHM_DateTime

In ESMP, the date and time is expressed as "YYYY-MM-DDThh:mmZ", which conforms with ISO 8601 UTC time zone. This date and time is without the seconds.

!# shows all attributes of YMDHM_DateTime.

Table !# - Attributes of ESMPDataTypes::YMDHM_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	The date and time as "YYYY-MM-DDThh:mmZ", which conforms with the ISO 8601 UTC time zone.

Enumerations

MessageTypeList

The coded type of a document. The message type describes the principal characteristic of a document. This enumeration is used in the XML instances based on IEC 62325.

Literal	Description	Code
Balance responsible schedule	A schedule which has been prepared by a balance responsible party providing planned schedule information.	A01
Allocated capacity schedule	A schedule which has been prepared by a capacity allocator providing allocated capacity.	A02
Balance area schedule	A schedule that provides the planned schedule information for a balance area.	A03
System Operator area schedule	A compilation of all external schedules concerning two System Operator areas or a connector concerning two System Operator of all balance responsible parties.	A04
Control block area schedule	A compilation of all the exchange programs of all control areas for one control block with all neighbouring control areas of a neighbouring control block.	A05
Coordination center area schedule	A compilation of the exchange programs of all exchange blocks divided into UCTE south and north.	A06
Intermediate confirmation report	An intermediate confirmation report that may be produced between final cutoffs.	A07
Final confirmation report	A final confirmation report that is produced after a final cutoff.	A08
Finalised schedule	A compilation of a set of schedules that have been finalized after a given cutoff.	A09
Regulation data report	A compilation of the time series employed on a given day to ensure the balance of the system.	A10
Aggregated energy data report	A compilation of the time series of all the meter readings or their equivalent for a given period.	A11
Imbalance report	The report containing the complete situation of a given period for a party and including the schedules, regulation data and actual or calculated readings.	A12
Interconnection Capacity	Document for cross-border capacity exchanges.	A13
Resource Provider Resource Schedule	A document providing the schedules for resource objects submitted by a resource provider.	A14
Acquiring System Operator Reserve Schedule	A document providing reserve purchases submitted by an Acquiring System Operator.	A15
Anomaly Report	A document providing anomaly information for the receiving party to correct.	A16
Acknowledgement Document	A document providing acknowledgement information.	A17
Confirmation report	A document providing confirmation information.	A18
Capacity for Resale Approved Capacity	A document providing information about capacity for resale.	A19

Transfer	A document to approve a capacity transfer.	A20
Capacity transfer notification	A document notifying a capacity transfer.	A21
Transmission rights portfolio	A document providing the portfolio of the transmission capacity rights of a market participant.	A22
Allocations	A document providing the capacity allocations for a border.	A23
Bid document	A document providing bid information.	A24
Allocation result document	A document providing the allocation results of an auction.	A25
Capacity document	A document providing capacity information.	A26
Rights document	A document providing transmission capacity rights information.	A27
Generation availability schedule	This document contains information related to energy availability.	A28
Cross border schedule	This document contains the cross border schedules for all the borders of a given country where energy is exchanged.	A30
Agreed capacity	The capacity agreed between parties.	A31
Proposed capacity	The capacity proposed for agreement between parties.	A32
System vertical load	The sum of all flows out of the transmission grid via directly connected transformers and lines to distribution grids and end consumers as known by the System Operator.	A33
Escalation document	A document which requesting the escalation of a situation.	A34
Trouble shooting document	A document providing trouble shooting information for the resolution of a problem.	A35
Deactivation document	A document providing deactivation information.	A36
Reserve tender document	The document that is used for the tendering for reserves within the ERRP process.	A37
Reserve Allocation Result Document	The document used to provide the results of a Reserve auction.	A38
SATCR activation	The document is used to provide the activation of reserves through the SATCR process.	A39
DATCR activation	The document is used to provide the activation of reserves through the DATCR process.	A40
Activation response	The document is used to provide a response to a request to activate reserves.	A41
Tender reduction	The document is used to provide information concerning the reduction of a previously submitted tender.	A42
MOL Document	The document is used to provide Merit Order List information.	A43
Price Document	The document is used to provide market price information.	A44
Measurement Value Document	The document is used to provide measurement information from measurement devices.	A45
SOAM Document	The document is used to provide system operator accounting data for matching.	A46
SOVA Document	The document is used to provide system operator validated accounting data.	A47
CCVA Document	The document is used to provide coordination center validated accounting data.	A48
Daily settlement document	The document is used to provide daily settlement information.	A49
Weekly settlement document	The document is used to provide weekly settlement information.	A50
Capacity Auction Specification Document	The document is used to provide capacity auction specification information.	A51
Market Coupling	The document is used to provide the results of a market coupling	A52

Results Document	auction.	
Outage publication Document	The document is used to provide the outage information for publication.	A53
Forced generation outage Document	A document providing information on forced generation outages.	A54
Summarised Market Schedule	A compilation of all external schedules concerning two Market Balance Areas of all balance responsible parties.	A55
Compensation Program Schedule	A schedule that provides the schedule information for the compensation of unintended deviation.	A56
Load Frequency Control Program Schedule	A schedule that provides the schedule information for the Load Frequency Control Program of a Control Area or a Control Block.	A57
Timeframe Independent Schedule	A compilation of all external Timeframe Independent Schedules concerning two System Operators.	A58
Information request	An information request being made concerning some specific information.	A59
status request for a position independently from a specific process	A status request concerning the position of an object independently of any ongoing processes.	A60
Estimated Net Transfer Capacity	The estimated net transfer capacity for a given border.	A61
Compensation rights	The capacity rights that have been allocated as compensation.	A62
Redispatch notice	A notice to confirm the actions agreed between System Operators to resolve a congestion situation through redispatch.	A63
Tender reduction response	A response to a tender reduction request that provides corrections to the initial document.	A64
System total load	Total load', including losses without power used for energy storage, is equal to generation deducted with exports, added with imports and deducted with power used for energy storage.	A65
Final MOL	A document providing the information concerning the situation of the MOL at the end of an activation period.	A66
Resource Provider Schedule for production/consumption	A document providing the schedules for production/consumption for resource objects submitted by a resource provider.	A67
Installed generation per type	A document providing the installed generation per generation type.	A68
Wind and solar forecast	A document providing the forecast of wind and solar generation.	A69
Load forecast margin	A document providing the load forecast margin for a period.	A70
Generation forecast	A document providing the generation forecast for a period.	A71
Reservoir filling information	A document providing information concerning the filling of reservoirs.	A72
Actual generation	A document providing the actual generation for a period.	A73
Wind and solar generation	A document providing the generation of wind and solar energy for a period.	A74
Actual generation per type	A document providing the actual generation per generation type for a period.	A75
Load unavailability	A document providing the unavailability of units providing load on the network.	A76
Production unavailability	A document providing the unavailability of production units providing energy to the network.	A77
Transmission unavailability	A document providing the unavailability of transmission capacity on the network.	A78
Offshore grid infrastructure unavailability	A document providing the unavailability of an offshore grid infrastructure to the network.	A79

Generation unavailability	A document providing the unavailability of generation units providing energy to the network.	A80
Contracted reserves	A document providing the reserves contracted for a period.	A81
Accepted offers	A document providing the offers of reserves that have been accepted for a period.	A82
Activated balancing quantities	A document providing the quantities of reserves that have been activated for balancing.	A83
Activated balancing prices	A document providing the prices of the reserves that have been activated for balancing.	A84
Imbalance prices	A document providing the prices of reserves due to imbalance for a period.	A85
Imbalance volume	A document providing the volume of the imbalance for a period.	A86
Financial situation	A document providing the financial situation for reserves.	A87
Cross border balancing	A document providing the cross border balancing requirements for a period.	A88
Contracted reserve prices	A document providing the price of reserves contracted for a period.	A89
Interconnection network expansion	A document providing information on the expansion of the interconnection network.	A90
Counter trade notice	A document providing information on counter trades for a period.	A91
Congestion costs	A document providing the cost of congestion for a period.	A92
DC link capacity	A document providing the DC links for a period.	A93
Non EU allocations	A document providing allocations made to non EU member states.	A94
Configuration document	A document providing configuration information.	A95
Redispatch activation document	A document enabling the activation of a redispatch notice.	A96
Detailed activation history document	A document enabling a detailed history of activations.	A97
Aggregated activation history document	A document enabling an aggregated history of activations.	A98
HVDC Link constraints	A document providing the information concerning the maximum and minimum active power flow through each link can limited.	A99
HVDC Configuration	A document providing the information concerning the power set point. The mode in which the HVDC is managed.	B01
HVDC Schedule	A document providing the information for operating DC links.	B02
EIC code request	A document providing the information requesting a new EIC code.	B03
EIC code information	A document providing EIC information in a central registry exchange or information to an EIC participant.	B04
EIC code publication	A document providing EIC publication information in a web site publication of a limited set of information.	B05
Critical network element determination	A document providing all the elements necessary for the capacity coordinator to determine the transfer capacity and the associated critical network elements.	B06
Critical network element publication	A document providing all the elements necessary for the market information aggregator and TSOs to know the critical network elements which limit the transfer capacity.	B07
Flow based domain	A document providing the capacity domain and its limits available for the TSO.	B08
Flow based domain publication	A document providing the capacity domain and its limits available for the market.	B09
Flow based domain market impact publication	A document providing the capacity domain and its impacts on the market to be published.	B10

Anonymized flow based parameters publication	A document providing all the relevant flow based parameters in case of flow based capacity allocation.	B11
Critical network element market impact publication	A document providing the critical network elements and its impacts on the market to be published.	B12
Weather document	An estimation or prediction in advance of the weather by analysis of meteorological data and the results of what actually happened with the weather.	B13
Energy prognosis document	A document to provide the prognosis of energy production/load for a given period.	B14
Network constraint document	A document providing the network constraint situations used for the load flow studies. A network constraint situation includes contingencies, monitored elements and remedial actions.	B15
Aggregated netted external market schedule document	A document used to report aggregated netted external market schedules for a given border.	B16
Aggregated netted external TSO schedule document	A document used to report aggregated netted external TSO schedules for a given border.	B17
Reporting status market document	A document used to report the status of aggregated netted external market schedules, aggregated netted external TSO schedules and compensation program schedules on a given border.	B18
Reporting information market document	A document used to report the information concerning aggregated netted external schedules, aggregated netted external market schedules, aggregated netted external TSO schedules, compensation program schedules, netted area position schedules and netted area AC position schedules to an interested party.	B19
Status request for a reporting information market document	A document requesting the provision of a reporting information document.	B20
Reserve need document	Used by a TSO to send its reserve needs.	B21
Generation and load shift keys document	A document providing the values of the generation and load shift keys to be used on network model.	B22
Offers to be activated	A document containing the outcome of the process, with the list of offers that are to be activated by the TSO concerned and the results for its reserve needs.	B23
Clearing price	A document containing the outcome of the process, with the clearing prices for a period.	B24
Security analysis report	A document providing a report on a performed security analysis.	B25
Aggregated netted external schedule document	A document used to report aggregated netted external schedules for a given border.	B26
External TSO schedule	A document used to report external TSO schedules for a given border or interconnector.	B27
Move of scheduled production	A document indication a move of scheduled production.	B28
PS&LC results document	A document providing Pole Splitting & Loss Calculation results.	B29
Notification data market document	A document used to notify data to any information receiver.	B30
Additional Constraint document	A document describing additional constraints for a capacity calculation process.	B31
Operational state document	A document used for exchanging operational states for grid assets.	B32
Published offered capacity	A document providing the most recent values of offered capacity.	B33
Market result document	Published prices and volumes	B34

Area Configuration document	A document containing the definition of areas.	B35
Area Composition document	A document containing the relations between areas, i.e which Metering Grid Areas a Bidding Zone composed of.	B36
Connected Areas document	A document containing which other areas an area is connected to i.e. which Metering Grid Areas a Metering Grid Area is connected to.	B37
Settlement document	A document providing settlement information.	B38
Imbalance prognosis document	A document to provide the prognosis of energy imbalances for a given period.	B39
Complete set of bids	Submission of complete set of bids. If there are existing bids, they should be replaced.	B40
Merged MOL notice	A notice providing information on the MOL merging process.	B41
K-factor document	A document providing K-factor values.	B42
Settlement coordination document	A document providing settlement information for coordination between different parties.	B43
Financial settlement document	A document providing financial settlement information.	B44
Bid availability document	A document providing the detailed reasons for changing the availability or volume of a bid.	B45
Resource capacity unit document	A document providing information about resource capacity units.	B46
Other market information	A document providing other market information.	B47
Permission document	A document that contains detailed information about permissions.	B48
Requested capacity	A document providing requested capacity (e.g. ATC, CZCL) value.	B49
Voltage control activation document	A document providing the number of generating units to activate for voltage control, by means of their operation in synchronous condenser mode.	B50
Transmission inventory document	A document providing a transmission inventory.	B51
Transmission inventory	A document providing statistics on the transmission grid.	B52

BusinessTypeList

The exact business nature identifying the principal characteristic of a time series.

Literal	Description	Code
Production	The nature of the business being described is production details.	A01
Internal trade	The nature of the business being described is internal trade details.	A02
External trade explicit capacity	The nature of the business being described is external trade details between two areas with limited capacity requiring a capacity agreement identification.	A03
Consumption	The nature of the business being described is consumption details.	A04
External trade total	The nature of the business being described is external trade total.	A05
External trade without explicit capacity	The nature of the business being described is external trade details between two areas without requiring capacity allocation information.	A06
Net Production / Consumption	Net production/consumption - where signed values will be used. With the following rules: In area=Out area, In party=Out party, + means production and - means consumption.	A07
Net internal trade	Net internal trade - where the direction from out party (seller) to in party (buyer) is positive and the opposite direction is negative (with minus signs).	A08
IPP (Independent	A time series concerning the production schedule from an IPP.	A09

Power Producer)		
Tertiary control	A time series concerning tertiary reserve.	A10
Primary control	A time series concerning primary reserve.	A11
Secondary control	A time series concerning secondary reserve.	A12
Load profile	A time series concerning a load profile as calculated by a metered data aggregator.	A13
Aggregated energy data	A time series concerning adjusted metered readings received from a metered data collector and aggregated and validated by a metered data aggregator.	A14
Losses	A time series concerning losses that have been calculated for a tieline or an area.	A15
Transits (CBT)	A time series concerning inter area transit flows determined for CBT requirements.	A16
Settlement deviation	A time series concerning the imbalance energy calculated by an imbalance settlement responsible.	A17
Technical constraint deviation	A time series defining the imbalance between schedules accepted by the system operator due to technical constraints and schedules declared by the balance responsible party.	A18
Balance energy deviation	A time series defining the imbalance between the schedule of a balance responsible party that has been corrected by the system operator after using balance energy bids and the schedule that was accepted by the system operator due to technical constraints.	A19
Imbalance volume	A time series defining the imbalance between the actual meter readings and the schedule of the balance responsible party corrected by the system operator after using balance energy bids.	A20
Unintended energy	A timeseries concerning the volume of an unintended cross-border exchange of energy.	A21
Frequency control	A time series concerning primary and secondary reserve.	A22
Balance management	A time series concerning energy balancing services.	A23
Total trade	A time series concerning the total of both the internal and external trades.	A24
General Capacity Information	A time series providing the total capacity available on a TSO border.	A25
Available transfer capacity (ATC)	Available transfer capacity for cross-border exchanges.	A26
Net transfer capacity (NTC)	Net transfer capacity for cross-border exchanges.	A27
Control Area Program	A time series providing the total exchanges between two TSOs (including the commercial transactions, the compensation program and the losses compensation program). Note this definition might change when UCTE brings forward its coding requirements.	A28
Already allocated capacity (AAC)	The already allocated capacity is the total amount of allocated transmission rights.	A29
Internal inter area trade	A trade that occurs between internal areas within a market balance area.	A30
Offered Capacity	The time series provides the offered capacity.	A31
Capacity transfer notification	The time series provides information concerning the notification of the transfer of capacity to another market participant.	A32
Authorised AAC	The time series in question provides the amount of transmission capacity	A33

	rights to be nominated.	
Capacity rights	The time series in question provides the capacity rights allocated for a given border.	A34
Minimum authorised AAC	The time series in question provides the minimum amount of transmission capacity rights to be nominated.	A35
Maximum authorised AAC	The time series in question provides the maximum amount of transmission capacity rights to be nominated.	A36
Installed generation	The time series in question provides the installed generation.	A37
Available generation	The time series in question provides the available generation.	A38
Interconnection Trade Responsible Designation	The Time series in question provides the designation of the ITR that may nominate the capacity in question.	A40
Released AAC	The already allocated capacity (AAC) that has been released for resale.	A41
Requested capacity (with price)	The time series in question provides information concerning the requested capacity including price information.	A42
Requested capacity (without price)	The time series in question provides information concerning the requested capacity but excludes price information.	A43
Compensation program	Compensation of unintentional deviation is performed by exporting to / importing from the interconnected system during the compensation period by means of schedules as calculated during the accounting of unintentional deviations.	A44
Schedule activated reserves	The cross border or internal reserves that are to be activated through schedule nomination.	A45
System Operator redispatching	The cross border redispatching between System Operators that are to be activated through schedule nomination.	A46
Market capacity price	The price of the capacity offered on a given market.	A47
Market capacity price differential	The difference between the price of capacity in a Market Balance Area receiving the capacity (In Area) and the price of capacity in a Market Balance Area providing the capacity (Out Area), i.e. In Area Price - Out Area price.	A48
Inflow	The volume of water that flows into a reservoir in a given interval.	A49
Water extraction	The volume of water that can be extracted from a reservoir in a given interval.	A50
Turbined water	The volume of water that can be turbined in a plant in a given interval.	A51
Water spillage	The volume of water that is not turbined going through the spillway in a given interval.	A52
Planned maintenance	Maintenance has been planned for the object in question with a forecast ending date.	A53
Unplanned outage	An unplanned outage has occurred on the object in question.	A54
Use it Or Sell it (UIOSI) pricing	The time series provides information on the capacity resold in the "use it or sell it" process and its corresponding price.	A55
Compensation for auction cancellation where capacity is for resale	The time series provides information on the compensation of the capacity for resale following an auction cancellation.	A56
Resale pricing	For each Physical Transmission Rights holder, this document contains the resold capacity and its corresponding price.	A57
Curtailed capacity compensation	The time series provides information to compensate a party when curtailment is applied on the capacity obtained in a previous auction, resale or transfer.	A58

Use it Or Sell it (UIOSI) compensation	The time series provides information on the compensation for the capacity following an auction cancellation.	A59
Minimum possible	The time series provides a schedule of minimum possible values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A60
Maximum available	The time series provides a schedule of maximum available values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A61
Spot price	The time series provides the market spot prices from an auction.	A62
Minimum ATC	The Available Transmission Capacity that must be guaranteed because of regulatory constraints.	A63
Meter Measurement data	The data as provided for a meter measurement source.	A64
Accounting Point Relevant data	The metered data that is to be considered relevant for accounting purposes.	A65
Energy flow	Energy flow information.	A66
Power plant energy Schedule	Energy flow scheduled for a power plant.	A67
Compensation Requirements for the compensation period	The time series provides the compensation requirements for a given compensation period.	A68
Market coupling results	The time series provides the results of a market coupling auction.	A69
Production, unavailable	Production capacity that normally would be available, but due to maintenance or similar is temporarily unavailable.	A70
Supplementary available generation	The supplementary generation that is available.	A71
Interruptible consumption	The consumption that may be interrupted on request.	A72
Summarised Market Balance Area Schedule	A time series providing the total exchanges based on commercial transactions between two Market Balance Areas.	A73
Load Frequency Control Program Schedule	A time series providing the schedule information for the Load Frequency Control Program.	A74
Timeframe Independent Schedule	A time series providing the total exchanges of Timeframe Independent Schedules between two System Operators.	A75
Consumption curtailment	A time series providing the amount of voluntary consumption curtailed by the energy supplier of an end-consumer.	A76
Production, dispatchable	The nature of the business being described is dispatchable production details, i.e. generation output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A77
Consumption, dispatchable	The nature of the business being described is dispatchable consumption details, i.e. consumption output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A78
Production, non-dispatchable	The nature of the business being described is non-dispatchable production details, i.e. generation output that cannot be modified by an activation order.	A79
Consumption, non-dispatchable	The nature of the business being described is non-dispatchable consumption details, i.e. consumption output that cannot be modified by an activation order.	A80
Total Transfer	The Total Transfer Capacity is the maximum exchange program between two areas compatible with operational security standards applicable at each	

Capacity (TTC)	system if future network conditions, generation and load patterns were perfectly known in advance.	A81
Mutual Emergency Assistance Service (MEAS)	The cross border Mutual Emergency Assistance Service between System Operators that are to be activated through schedule nomination.	A82
Auction cancelation	The time series covers auction cancellation right.	A83
Nomination curtailment	The time series covers nomination curtailment rights	A84
Internal redispatch	Redispatch to relieve Market Balance Area internal congestion.	A85
Control area balance energy	A sum of secondary, tertiary control as well as other energy that was used to balance a control area.	A86
Balancing energy price	Price of energy used to balance.	A87
Economised secondary reserve	The activated secondary reserve that had been economised due to pooled reserve management.	A88
Spinning reserve	The extra generating capacity that is available by increasing the production of generators that are already connected to the power system.	A89
Solar	The business being described concerns solar power.	A90
positive forecast margin	The business being described concerns a positive forecast margin.	A91
Negative forecast margin	The business being described concerns a negative forecast margin.	A92
Wind generation	The business being described concerns wind generation.	A93
Solar generation	The business being described concerns solar generation.	A94
Frequency containment reserve	The business being described concerns frequency containment reserve.	A95
Automatic frequency restoration reserve	The business being described concerns automatic frequency restoration reserve.	A96
Manual frequency restoration reserve	The business being described concerns manual frequency restoration reserve.	A97
Replacement reserve	The business being described concerns replacement reserve.	A98
Financial information	The business being described concerns financial information.	A99
Interconnector network evolution	The business being described concerns interconnector network evolution.	B01
Interconnector network dismantling	The business being described concerns interconnector network dismantling.	B02
Counter trade	The business being described concerns counter trades.	B03
Congestion costs	The business being described concerns congestion costs.	B04
Capacity allocated (including price)	The business being described concerns capacity allocation and includes price information.	B05
DC link constraint	The business being described concerns DC link constraints.	B06
Auction revenue	The business being described concerns auction revenue.	B07
Total nominated capacity	The business being described concerns the total nominated capacity.	B08

Net position	The business being described concerns net position.	B09
Congestion income	The business being described concerns congestion income.	B10
Production unit	The business being described concerns a production unit.	B11
Rounded market coupling results	Rounded outputs of the market coupling to be sent to TSOs and Market Participants.	B12
Allocation Revenue	The time series provides information on the revenue generated by the allocations.	B13
Production deviation	A time series concerning the imbalance energy between the metered production and the schedules calculated by an imbalance settlement responsible.	B14
Consumption deviation	A time series concerning the imbalance energy between metered consumption and the forecasted consumption calculated by an imbalance settlement responsible.	B15
Transmission asset	The business being described concerns a transmission asset.	B16
Consumption unit	The business being described concerns a consumption unit.	B17
In-feed ATC	Available Transfer Capacity at the in-feed side of a DC tieline.	B18
Out-feed ATC	Available Transfer Capacity at the out-feed side of a DC tieline.	B19
Balance up regulation price	A time series concerning balance regulation market prices for up regulation.	B20
Balance down regulation price	A time series concerning balance regulation market prices for down regulation.	B21
Main direction	A time series concerning the direction of balance regulations.	B22
Consumption imbalance price	A time series concerning imbalance prices for consumption.	B23
Production sales imbalance price	A time series concerning imbalance prices for production sales.	B24
Production purchase imbalance price	A time series concerning imbalance prices for production purchase.	B25
Average balance price between MBAs	A time series concerning the average prices between Market Balance Areas.	B26
Pumped	A time series concerning the electricity consumption related to pumping.	B27
Large installation consumption	A time series concerning consumption from large installation.	B28
Metering Grid Area (MGA) imbalance	A time series concerning imbalance between reported consumption, production and exchange in a Metering Grid Area.	B29
HVDC Link settings	The time series in question provides HVDC Link settings.	B30
Transmission Reliability Margin (TRM)	A time series concerning Transmission Reliability Margin (TRM).	B31
Imbalance component for a pool	This information is used to provide to a pool manager the combined imbalance of all the pool participants.	B32
Area Control error (ACE)	The sum of the instantaneous difference between the actual and the set-point value of the measured total power value and Control Program including Virtual Tie-Lines for the power interchange of a LFC Area or a LFC Block and the frequency bias given by the product of the K-Factor of the LFC Area or the LFC Block and the Frequency Deviation.	B33
Area Control	A time series concerning the Area Control Error after applying the imbalance	

Error after Imbalance Netting	netting energy correction.	B34
Implicit and explicit trade total	The sum of cross border schedules based on implicit and explicit trades including long term, yearly, monthly, weekly, daily processes.	B35
Production units own consumption	The consumption of one or more production units.	B36
Constraint situation	The timeseries describes the constraint situation for a given TimeInterval. A constraint situation can be: - composed of a list of network elements in outage associated for each outage to a list of network elements on which remedial actions have been carried out accordingly to contingency process - or it can be an external constraint.	B37
Initial domain	The timeseries describe the full flow based domain for a given TimeInterval. Critical network elements are displayed in details and their impact on the market is quantified.	B38
Flow based domain adjusted to long term schedules	The timeseries describe the full flow based domain for a given TimeInterval adjusted to the latest update of the schedules. Critical network elements are displayed in details and their impact on the market is quantified.	B39
Network element constraint	The timeSeries describes limiting elements which are overloaded.	B40
Calculation opposition (Red Flag)	The timeSeries describes a party who is opposed to the calculation result and imposes its transfer capacity value.	B41
Base case proportional shift key	The GSK or LSK are proportional to the base case generation or load.	B42
Proportional to participation factors shift key	The GSK or LSK are proportional to the participation factors.	B43
Proportional to the remaining capacity shift key	The GSK is proportional to the remaining available capacity.	B44
Merit order shift key	The GSK is proportional to a merit order list.	B45
Wind speed	The TimeSeries provides information on the wind speed.	B46
Wind direction	The TimeSeries provides information on the wind direction.	B47
Solar irradiance	The TimeSeries provides information on the power per unit area produced by the sun in the form of electromagnetic radiation.	B48
Air temperature	The TimeSeries provides information on the air temperature.	B49
Cloudiness	The TimeSeries provides information on the cloudiness, i.e. the level of coverage of the sky with clouds.	B50
Air humidity	The TimeSeries provides information on the level of humidity of the air.	B51
Atmospheric pressure	The TimeSeries provides information on the atmospheric pressure.	B52
Precipitation	The TimeSeries provides information on the amount of rain, snow, etc. that falls on the ground.	B53
Network constraint situation that constraints the market	The TimeSeries describes the network elements, that constraints the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	B54
Contingency	The TimeSeries describes the network elements part of the contingency to be	B55

	simulated for a given TimeInterval.	
Remedial Action	The TimeSeries describes a set of remedial actions for a given TimeInterval.	B56
Monitored Network Element	The TimeSeries describes the network elements to be monitored during the network load flow studies.	B57
Busbar	The TimeSeries describes the network elements that composed a busbar.	B58
Network Element	The TimeSeries describes network elements.	B59
SPS	The TimeSeries describes the network elements managed by a Special Protection System (automation).	B60
Aggregated netted external market schedule	The aggregated netted external market schedules for a given border.	B61
Aggregated netted external TSO schedule	The aggregated netted external TSO schedules for a given border.	B62
Aggregated netted external schedule	The aggregated netted external schedules for a given border.	B63
Netted area AC position	The AC position for a given area.	B64
Netted area position	The netted aggregation of all AC external schedules of an area plus the aggregated External Netted Schedules of related HVDC links of an area.	B65
Interconnection shift key	The shift key series describes the amount of power to be shifted from a border area.	B66
DC flow with losses	DC flow with losses refers to the values at the importing end of the DC line.	B67
DC flow without losses	DC flow without losses refers to the values at the exporting end of the DC line.	B68
minimum value of netted area position	That value which a netted area position must not fall below for a given area.	B69
maximum value of netted area position	That value which a netted area position must not exceed for a given optimisation area.	B70
maximum value of DC flow	That value which a balanced DC flow must not exceed for a given DC line on exporting end. When aligning DC flows CGMA algorithm will respect this constraint.	B71
minimum value of DC flow	That value which a balanced DC flow must not fall below for a given DC line on exporting end. Currently this business type is only included for consistency reasons. It is always set to 0. This constraint might, however, be used in future. When aligning DC flows the CGMA algorithm will respect this constraint.	B72
indicative AC flow	It is the hypothetical flow on the aggregate of all AC tie lines of an electrical border between two optimisation areas. It results from the adjustments to the preliminary netted area positions of all optimisation areas made by the CGMA algorithm. Indicative AC flows are an artefact of the CGMA algorithm, and do not correspond to physical flows	B73
Offer	The time series provides an offer to provide reserves.	B74
Need	The time series provides a requirement for reserves.	B75
Opportunity costs or benefits	The time series describes any opportunity costs or benefits.	B76
Financial compensation or penalties	The time series describes any financial compensation or penalties	B77
Global radiation	The total short-wave radiation from the Global radiation is the total short-wave radiation from the sky falling onto a horizontal surface on the ground. It includes both the direct solar radiation and the diffuse radiation resulting from reflected or scattered sunlight.	B78

Diffuse radiation	Radiation resulting from reflected or scattered sunlight.	B79
Direct solar radiation	Radiation resulting from direct sunlight	B80
Outage (OUT)	Outage process: Element is out of operation due to planned maintenance or due to an unplanned/forced outage. Outage may be used as a synonym for unavailability.	B81
Special switching state (SSS)	Outage Process: This state applies to grid elements which are in operation in an exceptional state (e.g. separated nodes operation).	B82
Testing (TEST)	Outage process: TESTING means any element status is possible - ON or OUT. This status applies either between first connection and final commissioning of the relevant asset, or directly following maintenance of the relevant asset.	B83
Auxiliary busbar operation	Outage process: Element is in operation but connected via auxiliary busbar	B84
Automatic reclosing	Outage process: Protection function Automatic reclosing is switched off for electric line	B85
Busbar protection	Protection function busbar protection is switched off	B86
Phase Shift Angle	The maximum phase shift angle allowed between two network elements.	B87
Base Case Network Situation	The TimeSeries describes the network elements to be taken into account to simulate a base case network situation during the network load flow studies, without any contingency.	B88
Inter-TSO assistance	Cross border assistance schedule between TSOs not interconnected directly.	B89
FlexibleNeed	The business type indicates that the need is optional.	B90
GLSK Limitation	A constraint related to a GLSK maximum or minimum limitation in the production or/and consumption shift.	B91
Capacity ramping limitation	A constraint related to a ramping limitation on the capacity offered at a given border.	B92
interconnector capacity	The maximum capacity that can be exchanged on an interconnector, excluding external factor on both ends.	B93
Must Run	A time series concerning must run generation.	B94
Procured capacity	An accepted offer of balancing capacity.	B95
Used capacity	The used cross-zonal balancing capacity.	B96
Estimated costs	Estimated costs of the process.	B97
Estimated benefits	Estimated benefits of the process.	B98
Load Shedding	A time series concerning a load shedding used to avoid failure of the power system.	B99
Remaining Capacity	A time series concerning the remaining capacity.	C01
Indicator of generation capacity adequacy	Indicator of adequacy, it indicates if there is final generation remaining capacity after SMTA calculation.	C02
Income from price divergence without congestions	The time series describes income due to price divergence without congestion between bidding zones.	C03
Push-button	The cross-border Push-button service between System Operators.	C04
Intertripping Emergency	The cross-border Intertripping service between System Operators.	C05

instruction	The cross-border Emergency instruction service between System Operators.	C06
Ramp management	The schedule resulting from cross-border Ramp management service between System Operators.	C07
Profile smoothing	The schedule resulting from cross-border Profile smoothing service between System Operators.	C08
Emergency reallocation deselection	The schedule resulting from cross-border Emergency reallocation deselection service between System Operators.	C09
SO-SO-trade	The generic cross border trade between System Operators.	C10
Production reduction	A time series providing the volume of production reduced by an energy provider / producer / supplier.	C11
Maximum power exchange	The timeseries provides the maximum admissible power flow between two bidding zones respecting operational security limits taking into account N-1 criterion.	C12
Maximum power exchange after remedial actions	The timeseries provides the maximum admissible power flow between two bidding zones after remedial actions.	C13
Network constraint situation that cannot limit the market	The TimeSeries describes the network elements, that cannot limit the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	C14
Flat participation for all generators or loads	Flat GSK factors of all generators or loads, independently of the size.	C15
Proportional to installed capacity of generators	Generators participate relative to their maximum (installed) capacity (MW).	C16
Market price and total volume	A time series concerning market price and total volume.	C17
Import price	A time series concerning import price (the volume-weighted price average of all accepted bids).	C18
Capacity allocated (excluding price)	The business being described concerns capacity allocation and excludes price information.	C19
Common Grid Model Equipment	The timeseries provides equipment related to the Common Grid Model (CGM).	C20
Exchanged balancing reserve capacity	The balancing reserve capacity exchanged between areas.	C21
Shared balancing reserve capacity	The balancing reserve capacity shared between areas.	C22
Share of reserve capacity	A time series concerning the share of reserve capacity.	C23
Actual reserve capacity	A timeseries concerning actual reserve capacity.	C24
K-factor	K-factor as stated in the SO GL Art. 2 (45). It is also known as Frequency Bias.	C25
Frequency Containment	FCR-N is a reserve that is automatically activated in both directions around a set point when the frequency varies between 50.10 Hz and 49.90 Hz after an	C26

Reserve-Normal (FCR-N)	imbalance.	
Frequency Containment Reserve-Disturbance (FCR-D)	FCR-D is a reserve that is automatically activated when the frequency falls below 49.90 Hz or rises above 50.1 Hz after an imbalance.	C27
Internal trade difference	A time series concerning internal trade difference, within an area, such as a Bidding Zone or Scheduling Area. The internal trade difference is the difference between trades reported from an out party (seller) and an in party (buyer).	C28
Small scale production	Production from small scale production plants.	C29
System price	The system price is an unconstrained market clearing reference price. It is calculated without any congestion restrictions by setting capacities to infinity.	C30
Wind gust	An increase in the speed of the wind lasting for a short period.	C31
Area imbalance	A time series concerning imbalance between planned consumption, production and exchange in an Area.	C32
Unintended energy price	A timeseries concerning the price of the unintended cross-border exchange of energy.	C33
Frequency containment process energy	A timeseries containing the volume of energy resulting from the frequency containment process.	C34
Frequency containment process energy price	A timeseries containing the energy price from the frequency containment process.	C35
Ramping period energy	A timeseries containing the volume of energy exchanged as a result of ramping between different ANES values.	C36
Ramping period energy price	A timeseries containing the price of the energy exchanged as a result of ramping between different ANES values.	C37
Frequency deviation	A timeseries concerning the difference between the actual and the nominal frequency of a synchronous area.	C38
Day-Ahead market price	A timeseries concerning Day-Ahead market prices.	C39
Conditional bid	Standard product bid that is conditional on bids submitted outside of common platform.	C40
Thermal limit	The current causing a given network element to work outside of the range of safe operating temperatures.	C41
Frequency Limit	A constraint related to the containment of frequency deviations within a given area.	C42
Voltage limit	The maximum or minimum permissible voltage within normal operation state of a given network element.	C43
Current limit	The maximum permissible current within normal operation state of a given network element.	C44
Short circuit current limit	The maximum permissible short-circuit current within normal operation state of a given network element.	C45
Dynamic stability limit	A maximum permissible load ensuring the control of oscillations in the grid and avoiding the loss of synchronism.	C46
Disconnection	A timeseries describing disconnection of a TSO from a common platform.	C47
Intended energy with positive price	A timeseries concerning the amount of intended energy with prices higher than zero (and including zero).	C48
Intended energy with negative price	A timeseries concerning the amount of intended energy with prices lower than zero (excluding zero).	C49
Decopuling	A time series describing decoupling of an area.	C50

Resource capacity unit	A timeseries containing information about resource capacity units.	C51
Resource entry capacity data	A timeseries containing the resource capacity that can be allocated to an eligible resource capacity operator from another area.	C52
Resource capacity obligation data	A timeseries containing the resource capacity operator obligation to guarantee delivery.	C53
Available energy	A timeseries concerning the available energy.	C54
Production curtailment	A timeseries concerning the curtailment of production.	C55
Rounding error	A timeseries describing a rounding error.	C56
Metered frequency	The timeseries provides information about metered frequency.	C57
Adjusted TTC to the nominal criteria	The exchange program between two areas which guarantees that the Margin Available for Cross-Zonal Trade (MACZT) fulfils the nominal criteria at least on the most limiting Critical Network Element with Contingency (CNEC) which limits the transfer capacity.	C58
Adjusted TTC to the nominal criteria with TSOs limitation	The exchange program between two areas which allows the Margin Available for Cross-Zonal Trade (MACZT) on, at least, the most limiting Critical Network Element with Contingency (CNEC) to get closer to the nominal criteria fulfilment with a limited impact on the rest of the network.	C59
Frequency deviation larger than standard deviation	Total time in which the absolute value of the instantaneous frequency deviation was larger than the standard frequency deviation.	C60
Frequency deviation larger than maximum deviation	Total time in which the absolute value of the instantaneous frequency deviation was larger than the maximum instantaneous frequency deviation.	C61
Frequency deviation not returned to 50%	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation as stated in SO GL (EU) regulation Art 131.1.a.vi.	C62
Frequency deviation not returned to restoration range	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation.	C63
Frequency deviation outside recovery range	Number of events for which the absolute value of the instantaneous frequency deviation was outside of the frequency recovery range.	C64
Frequency	A time series describing measurement frequency.	C65
Mean value	A time series describing mean values.	C66
Standard deviation	A time series describing standard deviation.	C67
Percentile	A time series describing percentiles.	C68
Measured frequency resolution	A time series describing the resolution of a measured frequency.	C69
Accuracy	A time series describing measurement accuracy.	C70
FRCE outside level 1 range	The number of time intervals in which the average value of the FRCE was outside the Level 1 FRCE range as stated in SO GL. (EU) regulation Art 131.1.b.i.	C71
FRCE outside level 2 range	The number of time intervals in which the average value of the FRCE was outside the Level 2 FRCE range as stated in SO GL (EU) regulation. Art 131.1.b.i.	C72

FRCE exceeded 60% of FRR capacity	The number of events for which the FRCE exceeded 60 % of the reserve capacity on FRR as stated in SO GL (EU) regulation Art 131.1.b.ii.	C73
FRCE exceeded steady state deviation	The number of events for which the absolute value of the FRCE exceeded the maximum steady-state frequency deviation.	C74
FRCE calculation and accuracy descriptor	A time series describing how FRCE is calculated and its accuracy.	C75
Forecasted capacity	A time series describing forecasted capacity.	C76
Minimum available capacity	A time series describing minimum available capacity.	C77
Average available capacity	A time series describing average available capacity.	C78
Maximum available capacity	A time series describing maximum available capacity.	C79
Frequency and accuracy descriptor	A time series describing how system frequency and accuracy are determined.	C80
Long-Term internal redispatch	A time series describing long-term redispatch to relieve Scheduling Area internal congestion.	C81
Other unavailability	This is an unplanned unavailability. Not considered by market participants as a planned maintenance.	C82
Faster than standard FAT	Bids that can support a ?Full Activation Time? (FAT) that is faster than standard FAT. Fast activation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C83
Faster than standard deactivation time	Fast deactivation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C84
Slower than standard FAT	Bids that can support a Full Activation Time (FAT) that is slower than standard FAT.	C85
Remedial action cost summary	A timeseries summarizing all incurred costs and/or revenues per party related to the activated remedial actions eligible to Cost Sharing.	C86
Settlement result	A timeseries representing the cost sharing settlement results between parties (i.e. which party will be paying and/or receiving money for the concerned period).	C87
Reserved cross zonal capacity	The reserved cross zonal capacity for system operator needs (EBGL art. 38(1) (b) and art. 41.).	C88
Energy reserves	A timeseries describing energy reserves.	C89
Combined dynamic constraint	The time series describes a combined dynamic constraint, being defined as a limit on the sum of power flows on a set of network elements or partial flows on a set of network elements for the purpose to respect dynamic stability limits.	C90
Anonymised constraint	The time series describes an anonymised constraint.	C91
Synchronous condenser activation	Activates a number of generators or other units in synchronous condenser mode.	C92
Permission administration details	A timeseries describing permission details such as authorisations, licenses or related regulatory requirements.	C93
Maximum capacity value		

of AC border flow	Maximum flow on an AC border between two areasÂ (capacity)	C94
Initial forecast value of AC border flow	Initial flow on an AC border between two areas (forecast)	C95

IndicatorTypeList

A boolean indicator to express Yes or No or True or False.

Literal	Description	Code
YES	A positive indication.	A01
NO	A negative indication.	A02

CurrencyTypeList

The coded identification of legal tender using ISO 4217 3 alpha codes.

Literal	Description	Code
Bosnian convertible marka	The Legal tender of Bosnia and Herzegovina.	BAM
Bulgarian lev	The Legal tender of Bulgaria.	BGN
Swiss Franc	The Legal tender of Switzerland.	CHF
Czech Koruna	The legal tender of the Czech Republic.	CZK
Danish Kroner	The Legal tender of Denmark.	DKK
EURO	The European legal tender.	EUR
Pound Sterling	The Legal tender of the United Kingdom.	GBP
Croatian kuna	The legal tender of Croatia.	HRK
Hungarian Forint	The Legal tender of Hungary.	HUF
Icelandic krona	The Legal tender of Iceland.	ISK
Albanian lek	The Legal tender of Albania.	LEK
Lithuanian litas	The Legal tender of Lithuania	LTL
Macedonian denar	The Legal tender of FYROM.	MKD
Norwegian Kroner	The Legal tender of Norway.	NOK
Polish zloty	The Legal tender of Poland.	PLN
Romanian Leu	The Legal tender of Romania.	RON
Serbian dinar	The Legal tender of Serbian Republic.	RSD
Saudi Arabian Riyal	The Legal tender of Saudi Arabia.	SAR
Swedish kroner	The Legal tender of Sweden.	SEK
Slovak Koruna	The legal tender of Slovakia.	SKK
New Turkish Lira	The Legal tender of Turkey.	TRY
US Dollar	The legal tender of the USA.	USD
Ukrainian hryvnia	The legal tender of Ukraine.	UAH

CodingSchemeTypeList

Codification scheme used to identify the coding scheme used for the set of coded values to identify specific objects.

Literal	Description	Code
EIC	The coding scheme is the Energy Identification Coding Scheme (EIC), maintained by ENTSO-E.	A01

Coding scheme which are part of GS1 system	The coding schemes maintained by GS1.	A10
Andorra National coding scheme	The National coding scheme of the country in question.	NAD
Albania National coding scheme	The National coding scheme of the country in question.	NAL
Armenia National coding scheme	The National coding scheme of the country in question.	NAM
Austria National coding scheme	The National coding scheme of the country in question.	NAT
Azerbaijan National coding scheme	The National coding scheme of the country in question.	NAZ
Bosnia and Herzegovina National coding scheme	The National coding scheme of the country in question.	NBA
Belgium National coding scheme	The National coding scheme of the country in question.	NBE
Bulgaria National coding scheme	The National coding scheme of the country in question.	NBG
Switzerland National coding scheme	The National coding scheme of the country in question.	NCH
Serbia and Montenegro National coding scheme	The National coding scheme of the country in question.	NCS
Czech Republic National coding scheme	The National coding scheme of the country in question.	NCZ
Germany National coding scheme	The National coding scheme of the country in question.	NDE
Denmark National coding scheme	The National coding scheme of the country in question.	NDK
Estonia National coding scheme	The National coding scheme of the country in question.	NEE
Spain National coding scheme	The National coding scheme of the country in question.	NES
Finland National coding scheme	The National coding scheme of the country in question.	NFI
France National coding scheme	The National coding scheme of the country in question.	NFR
United Kingdom National coding scheme	The National coding scheme of the country in question.	NGB
Georgia National coding scheme	The National coding scheme of the country in question.	NGE
Gibraltar National coding scheme	The National coding scheme of the country in question.	NGI
Greece National coding scheme	The National coding scheme of the country in question.	NGR
Croatia National coding scheme	The National coding scheme of the country in question.	NHR
Hungary National coding scheme	The National coding scheme of the country in question.	NHU
Ireland National coding scheme	The National coding scheme of the country in question.	NIE
Italy National coding scheme	The National coding scheme of the country in question.	NIT
Kyrgyzstan National coding scheme	The National coding scheme of the country in question.	NKG

Kazakhstan National coding scheme	The National coding scheme of the country in question.	NKZ
Liechtenstein National coding scheme	The National coding scheme of the country in question.	NLI
Lithuania National coding scheme	The National coding scheme of the country in question.	NLT
Luxembourg National coding scheme	The National coding scheme of the country in question.	NLU
Latvia National coding scheme	The National coding scheme of the country in question.	NLV
Morocco National coding scheme	The National coding scheme of the country in question.	NMA
Moldavia National coding scheme	The National coding scheme of the country in question.	NMD
Macedonia National coding scheme	The National coding scheme of the country in question.	NMK
Netherlands National coding scheme	The National coding scheme of the country in question.	NNL
Nordic Regional coding scheme	The coding scheme of the Nordic region which covers Denmark, Finland, Norway and Sweden.	NNN
Norway National coding scheme	The National coding scheme of the country in question.	NNO
Poland National coding scheme	The National coding scheme of the country in question.	NPL
Portugal National coding scheme	The National coding scheme of the country in question.	NPT
Romania National coding scheme	The National coding scheme of the country in question.	NRO
Russian Federation National coding scheme	The National coding scheme of the country in question.	NRU
Sweden National coding scheme	The National coding scheme of the country in question.	NSE
Slovenia National coding scheme	The National coding scheme of the country in question.	NSI
Slovakia National coding scheme	The National coding scheme of the country in question.	NSK
Turkey National coding scheme	The National coding scheme of the country in question.	NTR
Ukraine National coding scheme	The National coding scheme of the country in question.	NUA
CGM	The coding scheme used for Common Grid Model Exchange Standard (CGMES).	A02
Cyprus National coding scheme	The National coding scheme of the country in question.	NCY
ISO	The coding scheme for the preceding attribute is a code maintained by International Organization for Standardization (ISO), such as â€œISO 3166 *English country names and code elements*â€ .	A03

RoleTypeList

Identification of the role played by a party.

Literal	Description	Code
Trade	Refer to role model definitions in the ENTSO-E Harmonised Role Model	

responsible party	Document.	A01
Consumption responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A02
Combined power exchange (not to be used)	This role is no longer in the ENTSO-E Harmonised Role Model Document.	A03
System operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A04
Imbalance settlement responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A05
Production responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A06
Transmission capacity allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A07
Balance responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A08
Metered data aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A09
Billing agent	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A10
Market operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A11
Energy supplier	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A12
Consumer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A13
Control area operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A14
Control block operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A15
Coordination centre operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A16
Grid access provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A17
Grid operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A18
Meter administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A19
Party connected to grid	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A20
Producer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A21
Profile maintenance party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A22
Meter operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A23
Metered data collector	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A24
Metered data responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A25

Metering point administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A26
Resource Provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A27
Scheduling coordinator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A28
Capacity Trader	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A29
Interconnection Trade Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A30
Nomination Validator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A31
Market information aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that collects information from different sources and assembles it to provide a summary of the market.	A32
Information receiver	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party, not necessarily a market participant, which receives information about the market.	A33
Reserve Allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A34
MOL Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that Informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A35
Capacity Coordinator	A party, acting on behalf of the SOs involved, responsible for establishing a coordinated Offered Capacity and/or NTC and/or ATC between several Scheduling Areas.	A36
Reconciliation Accountable	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that is financially accountable for the reconciled volume of energy products for a profiled local metering point.	A37
Reconciliation Responsible	A party that is responsible for reconciling, within a metering grid area, the volumes used in the imbalance settlement process for profiled metering points and the actual metered quantities.	A38
Data provider	A party that is responsible for providing information to a central authority.	A39
Local Issuing Office (LIO)	A party that is responsible for operating a Local Issuing Office (LIO).	A40
Central Issuing Office (CIO)	A party that is responsible for operating a Central Issuing Office (CIO).	A41
EIC Participant	A party that participates in the EIC environment.	A42
Weather analyser	A party that analyses the current and forecast weather situation and establishes a prognosis of its impact on the renewable energy environment as well as the overall load.	A43
Regional Security Coordinator (RSC)	The RSC as defined in the System Operation guideline.	A44

Energy Service Company (ESCO)	A party offering energy-related services to the Party Connected to Grid, but not directly active in the energy value chain or the physical infrastructure itself. The ESCO may provide insight services as well as energy management services.	A45
Balancing Service Provider	A party with reserve-providing units or reserve-providing groups able to provide balancing services to LFC Operators.	A46
Energy trader	A party that is selling or buying energy.	A47
LFC Operator	A party that is responsible for the Load Frequency Control of its LFC Area or LFC Block.	A48
Transmission System Operator (TSO)	The Transmission System Operator (TSO) is responsible for the transport of electricity on the extra high-voltage and high-voltage interconnected system. This is a market participant and not a role in Harmonised Role Model.	A49
Distribution System Operator (DSO)	Distribution System Operator (DSO) is responsible for transport of electricity on high-voltage (optionally), medium-voltage and low-voltage distribution systems. This is a market participant and not a role in Harmonised Role Model.	A50
Resource Capacity Mechanism Operator	Resource capacity mechanism operator is the party responsible to operate the resource capacity mechanism in a member state. It can either be the TSO or an independent party.	A51
Resource aggregator	A party that aggregates resources for usage by a service provider for energy market services.	A52
Cost sharing calculator	A party responsible for the mapping of the costs between areas and parties.	A53
Settlement responsible	A party responsible for settling the costs.	A54
Coordinated Capacity Calculator	Coordinated Capacity Calculator is the entity or entities with the task of calculating transmission capacity, at regional level or above.	A55
Flexibility service provider	A party that offers flexibility services based on acquired (aggregated) resources.	A56
Data access provider	A party responsible for facilitating access to data.	A57
Metered data administrator	A party responsible for storing and distributing validated measured data.	A58
Permission administrator	A party responsible for administrating a register of data access permissions for a set of metering points, making this information available to final customers and eligible parties in the sector, on request.	A59

UnitOfMeasureTypeList

(synonym MeasurementUnit) The unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

Literal	Description	Code
ampere	The unit of electrical current in the International system of Units (SI) equivalent to one Coulomb per second.	AMP
megavolt-ampere	MVA unit as per UN/CEFACT recommendation 20.	MVA
One	A unit for dimensionless quantities, also called quantities of dimension one.	C62
cubic hectometres	A unit of volume equal to one million cubic metres.	HMQ
kilovolt	kV unit as per UN/CEFACT recommendation 20.	KVT
kilometre	km unit as per UN/CEFACT recommendation 20.	KMT
kilovolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1. The unity power factor is expressed in	KVR

	thousands of a volt ampere reactive.	
gigawatt hour	GWh unit as per UN/CEFACT recommendation 20.	GWH
kilowatt hour	A total amount of electrical energy transferred or consumed in one hour.	KWH
kilowatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in thousands of a watt.	KWT
megavolt ampere reactive hours	Total amount of reactive power across a power system.	MAH
megavolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1.	MAR
megawatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in millions of a watt.	MAW
cubic metres per second	The volume flow rate of cubic metre per second.	MQS
cubic metre	A Cubic metre.	MTQ
metre	The length of a metre.	MTR
megawatt hours	The total amount of bulk energy transferred or consumed.	MWH
percent	A unit of proportion equal to 0.01.	P1
gigawatt	GW unit as per UN/CEFACT recommendation 20.	A90
watt	The watt is the International System of Units (SI) standard unit of power (energy per unit time), the equivalent of one joule per second.	WTT
meter per second	A unit of measurement of the speed expressed in m/s.	MTS
degree (unit of angle)	A unit of measurement of angles expressed in a 0 to 360 degree gradient.	DD
watt per square meter	A unit of measurement of the density of heat flow rate expressed in watt per square meter.	D54
Celsius	A unit of measurement of temperature expressed in degree Celsius.	CEL
OKTA unit	A unit of measurement of the cloudiness expressed in OKTA or OCTA, i.e. A unit of count defining the number of eighth-parts as a measure of the celestial dome cloud coverage.	A59
hectopascal	A unit of measurement of the pressure expressed in hectopascal.	A97
millimeter	A unit of measurement of length expressed in millimeter.	MMT
K (Kelvin)	Temperature unit refer ISO 80000-5 (Quantities and units, Part 5: Thermodynamics).	KEL
Hertz	HTZ unit as per UN/CEFACT recommendation 20.	HTZ
Megawatt per Hertz	A unit of energy expressed as the load change in million watts that will cause a frequency shift of one hertz.	E08
millihertz	A unit of frequency equal to 0.001 cycle per second.	MTZ
minute	A period of time equal to sixty seconds.	MIN
second	A period of time equal to one second.	SEC

Compounds

ESMP_DateTimeInterval

This datatype enables to express the start date and time, and the end date and time of a time interval with a specific pattern. This pattern is the YYYY-MM-DDThh:mmZ.

!# shows all attributes of ESMP_DateTimeInterval.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTimeInterval.

Attribute name	mult.	Attribute type	Description
start (Compound)	1..1	YMDHM DateTime	The start date and time of the interval with a minute resolution.
end (Compound)	1..1	YMDHM DateTime	The end date and time of the interval with a minute resolution.

Primitives

String

A string consisting of a sequence of 8 bit characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

DateTime

Date and time as "YYYY-MM-DDThh:mm:ss.sssZ", which conforms with ISO 8601 UTC time zone.
Note: the time within ESMP is expressed in UTC.

Integer

An integer number. The range is unspecified and not limited.

Decimal

Decimal is the base-10 notational system for representing real numbers.

Duration

Duration as "PnYnMnDTnHnMnS" which conforms to ISO 8601, where nY expresses a number ofyears, nM a number of months, nD a number of days. The letter T separates the date expression from the time expression and, after it, nH identifies a number of hours, nM a number of minutes and nS a number of seconds. The number of seconds could be expressed as a decimal number, but all other numbers are integers.